SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE CODE: CSA15

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 25

Dave's Taxonomy: Imitation (Level 1) Question
Count: 5

Code of Conduct :CSA1514

I P.B.VAMSI KRISHNA [192525147] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

P.*B.VAMSI*

Signature of the student:

Assessment Tasks

1) Define cloud computing and explain any four characterstics that make it different form

Definition :-

cloud computing is the delivery of computing service such as servers, storage, database, networking, and sofware over the internet on demand.

Four characterstics of cloud computing :

1) On demand self service :- users can access computing Resource whenever required without contacting the service provider.

Example: creating a virtual Machine in AWS

2) Broad network access :- cloud services are accessible through the internet using computers, tablets and smartphones.

3) Resource Pooling :- computing resources are shared among Multiple users using a Multiple tenant Model.

4) Rapid elasticity :- Resources can be increased or decreased quickly according to workload requirments.

2) Trace the evolution of cloud computing for hardware evolution to internet software evolution and show how these changes enabled model cloud platform.

cloud computing envolved through several important technological developments.

1) Mainframe computer: Early computing used large, centralized computers shared by multiple users through terminals.

2) Personal computer: computer became smaller and more affordable, allowing individuals and organizations to perform local computing.

3) client server computing: Processing the data were shared between client computer and centralized servers.

4) Virtualization :- Multiple virtual machine cloud run a single physical server improving Resource utilization and Reducing cost.

5) Internet development: High speed development internet enabled users to access application and computing Resource remotely.

6) web services and cloud platforms: web technologies automation, and distributed computing enabled services to be deliverd over the internet.

Explain the Role of sever virtualization in cloud computing

Server virtualization: It is a technology that divides one physical server into multiple independent virtual Machine using safware called Hypervisior

Role of server virtualization :-

- improves hardware utilization by running Multiple Machines on one servea.
- Reduce hardware utilization Naintainence cost.
- Enables ferible allocation of CPU, Nemory and storage.

Difference b/w virtualization and cloud computing :

VIRTUALIZATION	CLOUD COMPUTING
* A technology that creates virtual Resources	* A Model for delivering computing services over a netw
* Uses hypervisior to create virtual Machine	* Uses virtualization, automatio networking, and management tool
* May operate with a Private data center	* Provides on demand acces to services and Resources
* Example * VMware ESXi	* Example: * Amazon EC2

4) Discuss the role of data centers in cloud services

A data center is a facility containing servers, storage systems, networking equipments and supporting infrastructure used to store, process and manage data.

Role of data centers in cloud computing:

1) Data storage: store large amount of users and application data.

2) Processing: Provide CPU and memory resources to run application and virtual Machine.

3) Networking: connects servers to the internet and transfer data between users and cloud services.

4) Availability: Use backups systems, redundant Equipments, and disaster recovery Mechanisms to Minimize service interruptions.

5) Security: Protect data through access controls, Monitoring and physical security.

6) Scalability: Allow cloud providers to Expand computing capacity according to customer demand.

Example: Aws operates data Centers around the word to provide computing, storage, and data base services to customers.

Difference laas, Paas, and xaas with one suitable real world example for each model.

cloud computing provides different service models based on the level of control and management offered to users.

Service.NO	FULL FORM	DESCRIPTION	EXAMPLE
laas	infrastructure as a services	Provides virtual machines storage and networking where Manage the os and application	AMAZON EC2
Paas	Platform as a service	Provides a Platform and development tools to build and deploy application without managing the underlying infrastructure	Google app Engine
SaaS	software as a service	Provides Ready to use software through the internet	G-Mail
Xaas	Everything as a service	Delivers ready to use various IT Resources and capabilities as service over a network	Sales force CRM as a service.

Key difference:

:) laas : users have greater control over infrastructure & os

:) Paas : Developers focus mainly on application development

:) Saas : users access complete software without managing its infrastructure.

:) xaas : A board term covering many services delivered as needed storage, security and database.

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks: